

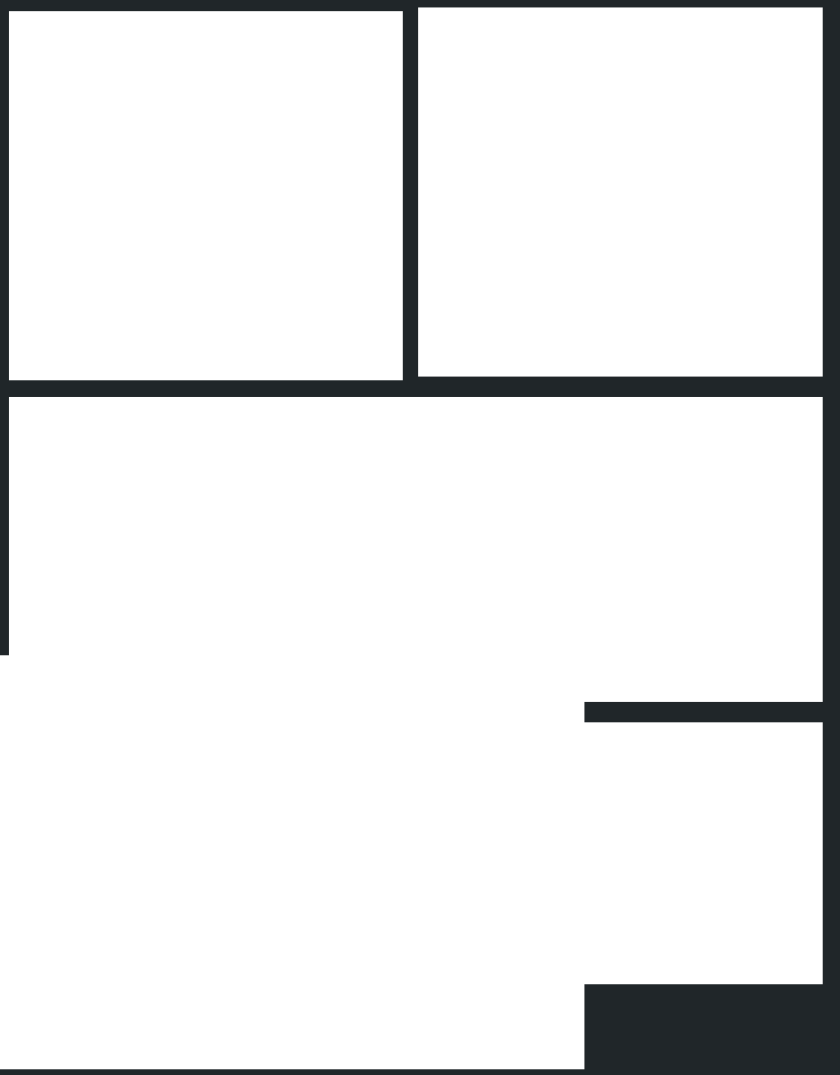
Reimagining the future of smart cities

Presenter:
DJ: Fengji
June 10, 2022

DESCRIPTION

B2B Tech Startup working towards decarbonization of commercial buildings





- Launched on July 5 2021 in Singapore
- Deep tech startup focused on machine learning innovation
- International team based in Singapore, Canada, US, and Europe
- Domain expertise in civil engineering, construction, and cyber physical systems
- Team background in computer science, engineering, finance, and clean tech

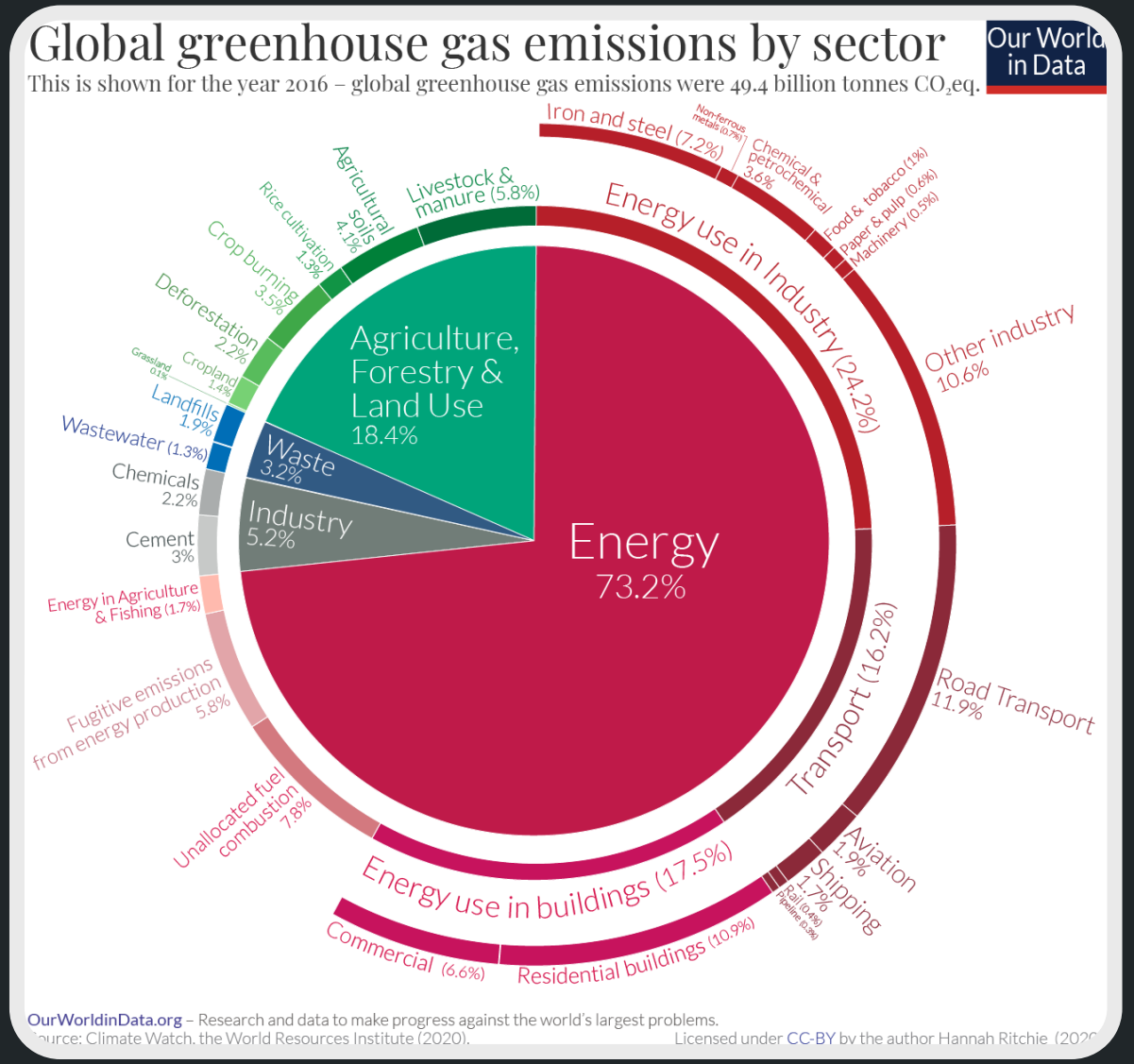
SUSTAINABILITY

Envisioning smart cities through data science and AI



Cloud Program

Google for Startups



[redacted] **hold significant ts**

Commercial Buildings: 6.6%

Road: 11.9%

Aviation: 1.9%

Shipping: 1.7%

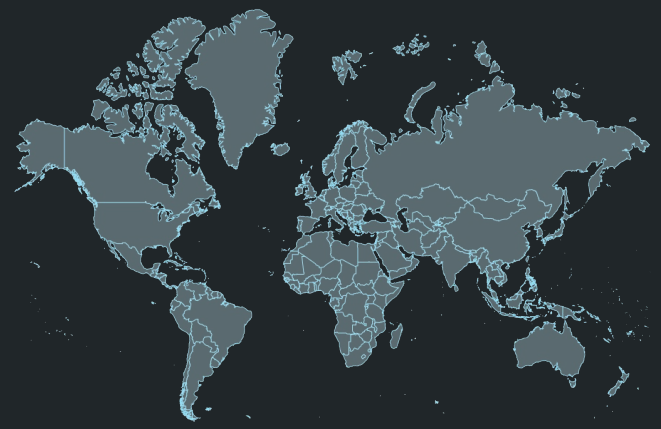
Transportation: 16.2%

Industry: 24.2%

~ 50% of global CO_2 emissions

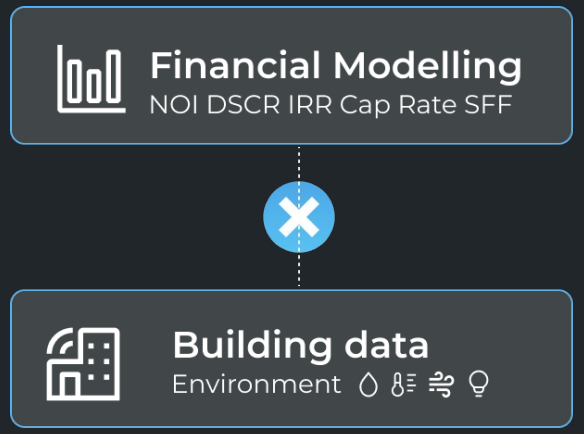
PROBLEM STATEMENT

Decarbonization solutions for Commercial Buildings are missing key items



01

Climate Targets not Fully Included



02

Financial Modelling disconnected from Building Data



03

Hard to Find Analytical Expertise

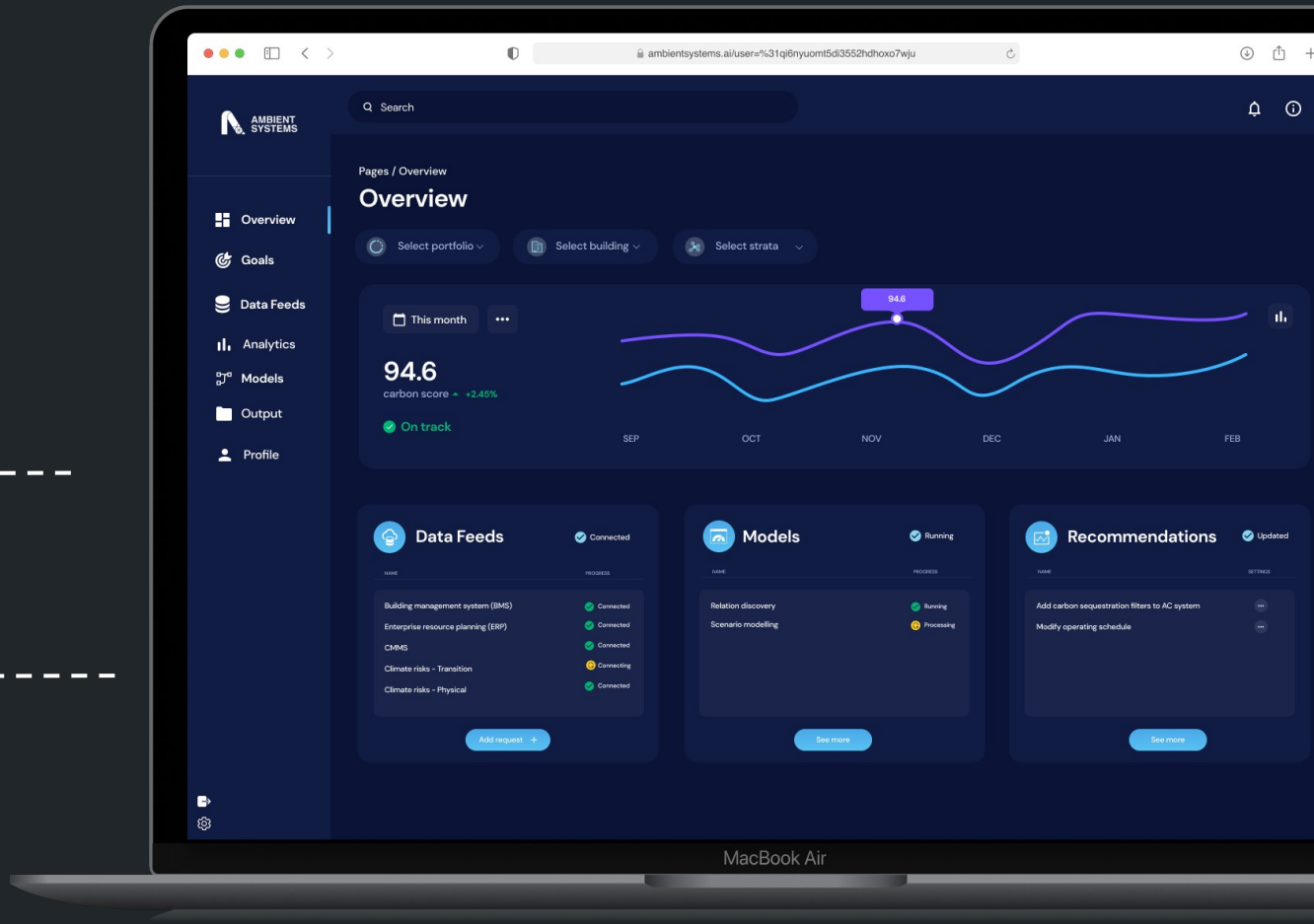
BUSINESS USE CASE

All-in-one Decarbonization Platform for Real Estate – powered by the Cloud & Data Science

AUTOMATE LIFE CYCLE ASSESSMENT TO QUANTIFY CARBON FOOTPRINT

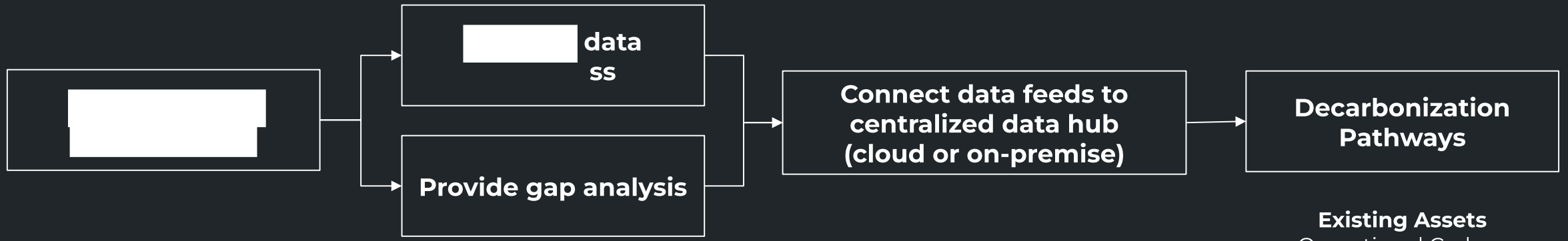
IDENTIFY NET ZERO PATHWAYS ACROSS DESIGN & OPERATIONS

DELIVER INSIGHTS & REPORTS TO STAKEHOLDERS



IMPLEMENTATION

Ambient delivers net zero pathways directly to front-line service teams & decision makers



1.
2. Requirements
3. Minimize energy intensity of existing services & products

Technical
CMMS
BMS/BIM
BOM

Risk
Energy Prices
Climate Risks
Labour & Materials Costs

Operational
Electricity
Gas & Fuel
Water

Existing Assets
Operational Carbon
(51% of emissions)

- Demand Shaping
- Electrification
- Efficiency
- DERs Deployment

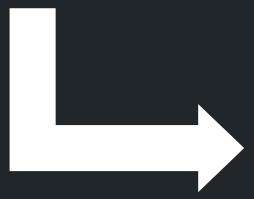
New Assets
Embodied Carbon
(49% of emissions)

- Passive Design
- Material Alternatives
- Efficient Design
- Construction Electrification
- Materials Routing
- Design for Recycling

HOW DOES IT WORK

Machine learning module provides recommendations based on operational data

Date/Time	Electricity:Facility [kW](Hourly)	Fans:Electricity [kW] (Hourly)	Cooling:Electricity [kW](Hourly)	Heating:Electricity [kW](Hourly)	InteriorLights:Electricity [kW](Hourly)	InteriorEquipment:Electricity [kW] (Hourly)	Gas:Facility [kW] (Hourly)	Heating:Gas [kW] (Hourly)	InteriorEquipment: Gas [kW](Hourly)	Water Heater:WaterSystems:Gas [kW] (Hourly)
01/01 01:00:00	174.2145729	32.75032584	48.42562787	0	34.43116061	33.10218689	154.7941419	28.41378994	3.33212	123.0482319
01/01 02:00:00	168.3524558	32.75032584	42.55659868	0	34.43116061	33.10218689	116.5007823	24.8622231	3.33212	88.30643925
01/01 03:00:00	146.3612177	32.75032584	38.07911969	0	19.46627941	30.61066878	119.1753379	22.2889279	3.33212	93.55428996
01/01 04:00:00	144.0482316	32.75032584	35.77850208	0	19.46627941	30.61066878	117.909866	20.51268543	3.33212	94.06506054
01/01 05:00:00	151.1611995	32.75032584	36.3191083	0	19.46627941	30.61066878	145.3588048	19.15022631	3.33212	122.8764585
01/01 06:00:00	165.1205174	32.75032584	38.04378529	0	19.46627941	30.61066878	180.6108081	18.307706	8.3303	153.9728021
01/01 07:00:00	[redacted]	35.03194464	45.96961404	0	42.69180509	54.3732441	[redacted]	17.7994769	16.6606	[redacted]
01/01 08:00:00	[redacted]	37.33275959	72.0152778	0	49.39604182	88.5934453	[redacted]	92.41443235	24.9909	[redacted]
01/01 09:00:00	[redacted]	37.31356344	102.6156863	0	57.6566863	112.0301055	[redacted]	96.21492711	33.3212	[redacted]
01/01 10:00:00	[redacted]	37.88787676	102.6893755	0	50.95244957	81.16842718	[redacted]	42.92042145	66.8089	[redacted]



Demand Shaping | Electrification | Efficiency | DERs Deployment

Ambient Systems

ambient-systems.web.app/Dashboard/Output/Recommendation

AMBIENT SYSTEMS
Climate At Your Service

Search...

- Overview
- Goals
- Data Engineering
- Data Visualization
- Machine Learning
- Output
- Profile

Ambient / Output / Recommendation

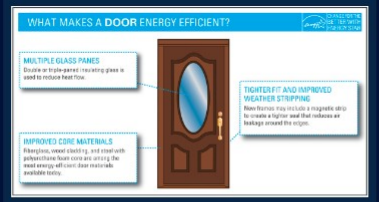
Recommendation

Select Portfolio | Select Building | Select Strata

Top Recommendations:

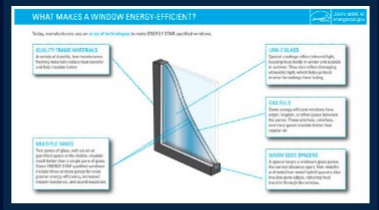
Replace your sliding glass doors with ENERGY STAR Most Efficient models:

- U-Factor 1.22 W/m²K or Energy Rating of 34 or more
- Up to 89% more efficient than standard sliding glass doors
- See this link: iconic-essence



Replace your windows with ENERGY STAR Most Efficient models:

- U-Factor of 1.05 W/m²K or less or Energy Rating of 40 or more
- up to 79% more efficient than standard windows
- See this link: vertical_slider_windows



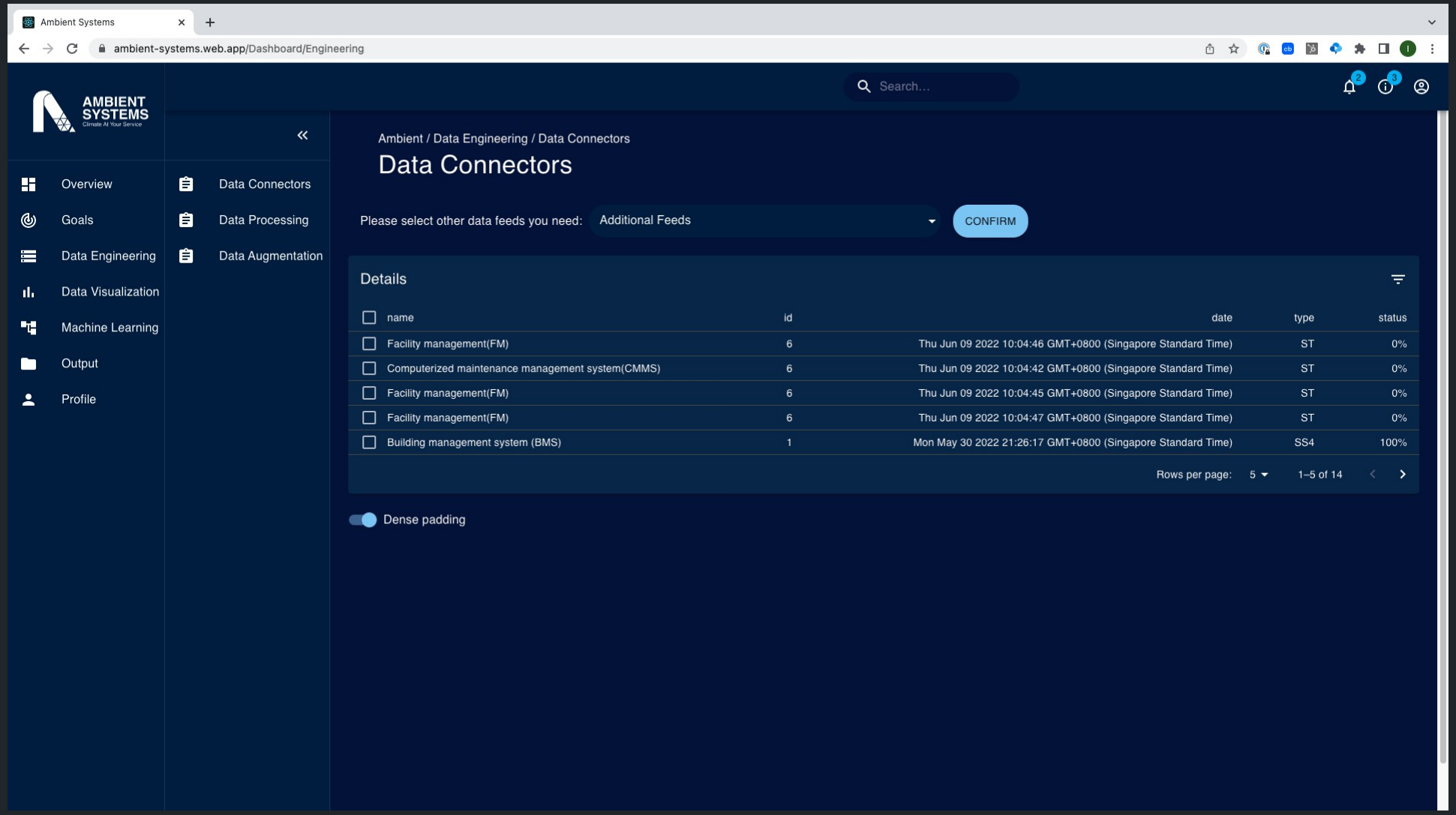
Replace your dishwasher with ENERGY STAR Most Efficient models:

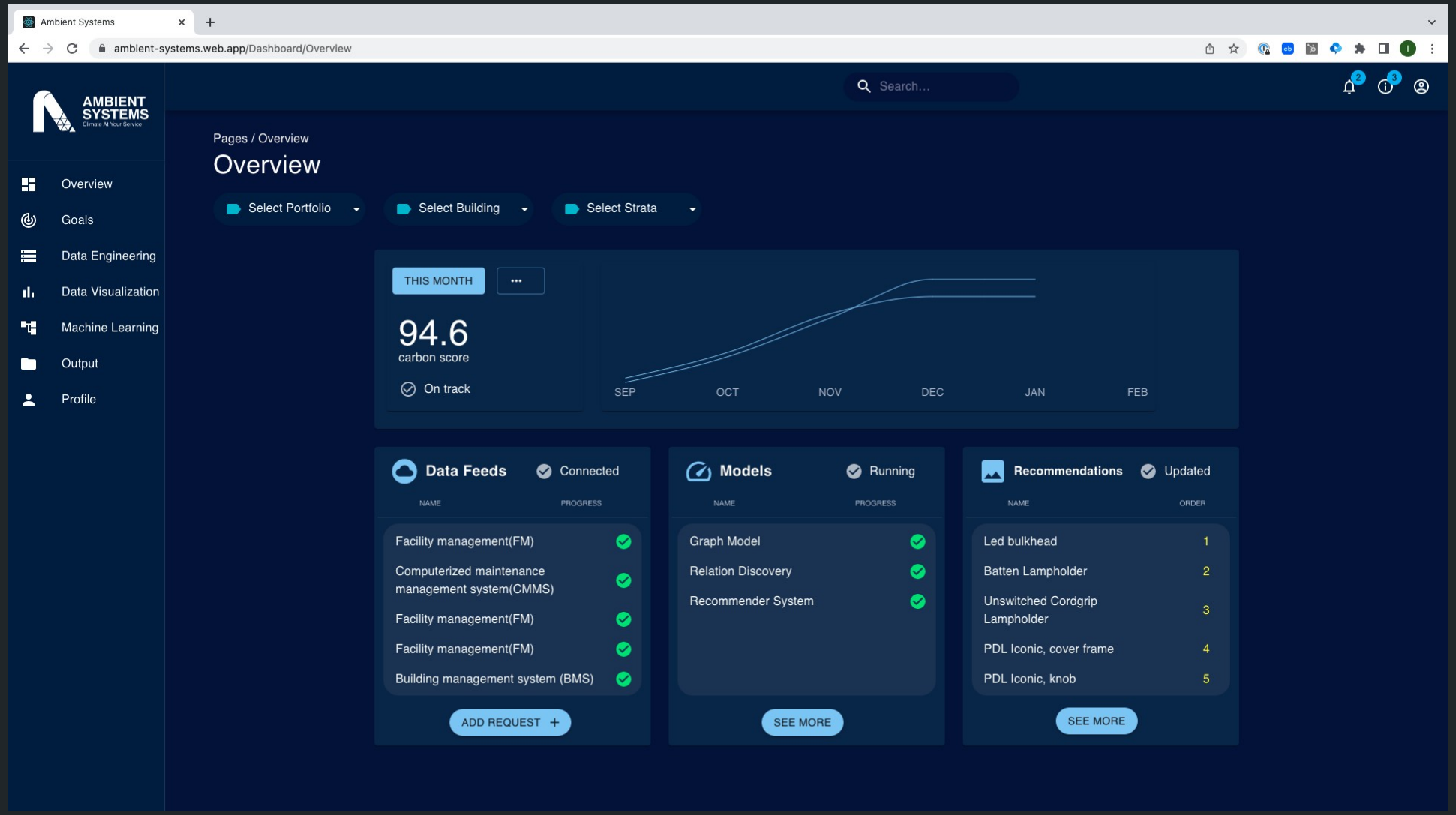
- 200 kWh / year or 0.8 to 0.9 kWh/ load, about 10-13 litres of water per use
- uses 12% less energy and 30% less water on average, than a standard model
- See this link:me-certified-dishwashers



Results in Details:

Product	Rank Level	Improved Carbon Score
Led bulkhead - 20W - 1100 lm - 4000K - IP54 body - rectangle	rank1	4.7
Batten Lampholder, Lighting, 230/240VAC, 150W	rank2	4.7
Unswitched Cordgrip Lampholder- 1 phase - 3A - 250V - 150W	rank3	4.3
PDL Iconic, cover frame, 6 switches, vertical	rank4	4.3
PDL Iconic, knob for cover frame, Iconic Essence	rank5	3.9
PDL Iconic Essence Single Power Point Skin with 1 extra switch, Horizontal Mount, 250V, 10A	rank6	3.6
PDL Iconic, cover frame, 5 switches, vertical	rank7	3.3


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Machine Learning
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Data Connectors
Data Processing
Data Augmentation
Ambient / Data Engineering / Data Connectors
Data Connectors
Please select other data feeds you need:
Additional Feeds
CONFIRM
Details
name
id
date
type
status
Facility management(FM) 6 Thu Jun 09 2022 10:04:46 GMT+0800 (Singapore Standard Time) ST 0%
Computerized maintenance management system(CMMS) 6 Thu Jun 09 2022 10:04:42 GMT+0800 (Singapore Standard Time) ST 0%
Facility management(FM) 6 Thu Jun 09 2022 10:04:45 GMT+0800 (Singapore Standard Time) ST 0%
Facility management(FM) 6 Thu Jun 09 2022 10:04:47 GMT+0800 (Singapore Standard Time) ST 0%
Building management system (BMS) 1 Mon May 30 2022 21:26:17 GMT+0800 (Singapore Standard Time) SS4 100%
Rows per page: 5
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Dense padding


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Select Portfolio
Select Building
Select Strata
THIS MONTH
94.6
carbon score
On track
SEP
OCT
NOV
DEC
JAN
FEB
Data Feeds
Connected
NAME
PROGRESS
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Computerized maintenance management system(CMMS)
Facility management(FM)
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Running
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PROGRESS
Graph Model
Relation Discovery
Recommender System
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Updated
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Batten Lampholder 2
Unswitched Cordgrip Lampholder 3
PDL Iconic, cover frame 4
PDL Iconic, knob 5
SEE MORE

Demo Video Link

Create a Free Demo Account at:

ARCHITECTURE DIAGRAM

Proprietary ML Models delivered through GCP services

